Program 1.1 Illustration of First Java Program

```
class Prg1 {
public static void main(String args[ ])
{
 System.out.println(“Hello World!!!”);
}
}
```

The output of the program will now be displayed as –

Hello World!!!

Program 1.2 Illustrating usage of variables

```
class Prg2 {
public static void main(String[] args)
{
 int n; n=25;
 System.out.println(“The value of n is: “ + n); n= n*3;
System.out.print(“The current value of n is: “ );
System.out.println(n);
}}
```

The output will be –

The value of n is: 25 The current value of n is: 75

Program 1.3 Illustration of if statement

```
class IfSample {
public static void main(String args[]) {
int x, y; x = 10; y = 20;
 if(x < y)
        System.out.println("x is less than y");
x = x * 2;
if(x == y)
        System.out.println("x now equal to y");
x = x * 2;
if(x > y)
        System.out.println("x now greater than y");
if(x == y)
        System.out.println("you won't see this");
 }
}
```

The output would be –

```
x is less than y
x now equal to y
x now greater than y
```

Program 1.4 Illustration of for statement

```
class ForTest
{
        public static void main(String args[])
        {
            int x;

            for(x = 0; x<5;x=x+1)
            System.out.println("This is x: " + x);
        }
}
```

This program generates the following output:

```
This is x: 0
This is x: 1
This is x: 2
This is x: 3
This is x: 4
```

Program 1.5: Program to illustrate need for long data type

```
class Light
{
        public static void main(String args[ ])
        {
```

```
        int lightspeed;
        long days, seconds, distance;
        lightspeed = 186000;
        days = 1000;

        seconds = days * 24 * 60 * 60;
        distance = lightspeed * seconds;
        System.out.print("In " + days);
         System.out.print(" days light will travel about ");
         System.out.println(distance + " miles.");
        }
}
```

The output will be –

In 1000 days light will travel about 16070400000000 miles**.**

Program 1.6 Finding area of a circle

```
class Area
{
        public static void main(String args[])
        {
         double pi, r, a;
         r = 10.8;
         pi = 3.1416;
         a = pi * r * r;
        System.out.println("Area of circle is " + a);
```

```
        }
}
```

The output would be –

Area of circle is 366.436224

Program 1.7 Demonstration of char data type

```
class CharDemo
{
        public static void main(String args[])
        {
                char ch1=88, ch2='Y';
                System.out.print("ch1 and ch2: ");
                System.out.println(ch1 + " " + ch2);
                ch1++; //increment in ASCII (even Unicode) value
                System.out.println("ch1 now contains "+ch1);
                --ch2; //decrement in ASCII (even Unicode) value
                System.out.println("ch2 now contains "+ch2);

                /*      ch1=35;
                        ch2=30;
                        char ch3;
                        ch3=ch1+ch2; //Error
                        */
                ch2='6'+'A'; //valid
                System.out.println("ch2 nowcontains "+ch2);
                }
```

}

The output would be –

ch1 and ch2: X Y

ch1 now contains Y

ch2 now contains X

ch2 now contains w

Program 1.8 Demonstration of Boolean data type

```
class BoolDemo
{
        public static void main(String args[])
        {
                boolean b = false;
                System.out.println("b is " + b);
                b = true;
                System.out.println("b is " + b);
                if(b)
                        System.out.println("True block");
                b = false;
                if(b)
                        System.out.println("False Block will not be executed");
                b=(3<5);
                System.out.println("3<5 is " +b);
            }
        }
```

The output would be –

b is false

b is true

True block

3<5 is true

Program 1.9 Demonstration of scope of variables

```
class Scope
{
	public static void main(String args[])
	{
		int x=10,i;			// x and i are local to main()
		if(x == 10)
		{
			int y = 20;		// y is local to this block
			System.out.println("x and y: " + x + " " + y);
			x = y * 2;
		}
			// y = 100;		//y cannot be accessed here
		System.out.println("x is " + x);
		for(i=0;i<3;i++)
		{
			int a=3; // a is local to this block
			System.out.println("a is " + a);
			a++;
		}
	}
```

```
}
```

The output would be –

x and y: 10 20

x is 40

a is 3

a is 3

a is 3

Program 1.10 Illustration of type conversion

```
class Conversion
    {
        public static void main(String args[])
        {
                byte b;
                int i = 257;
                double d = 323.142;
                System.out.println("\nConversion of int to byte.");
                 b = (byte) i;
                System.out.println("i and b " + i + " " + b);
                System.out.println("\nConversion of double to int.");
                i = (int) d;
                System.out.println("d and i " + d + " " + i);
                System.out.println("\nConversion of double to byte.");
                b = (byte) d;
                System.out.println("d and b " + d + " " + b);
            }
    }
```

The output would be –

```
Conversion of int to byte:        i = 257          b = 1
Conversion of double to int:     d = 323.142   i = 323
Conversion of double to byte: d = 323.142     b = 67
```

Program 1.11 Demonstration of type promotions

```
class TypePromo
        {
        public static void main(String args[])
        {
                        byte b = 42;
                        char c = 'a';
                        short s = 1024;
                        int i = 50000;
                        float f = 5.67f;
                         double d = .1234;
                        double result = (f * b) + (i / c) - (d * s);
                         System.out.println("result = " + result);
                }
        }
```

The output would be –

result = 626.7784146484375

Program 1.12 Demonstration of 2-d array

```
class TwoDArray
    {
        public static void main(String args[])
        {
                int twoD[][]= new int[3][4];
                int i, j;
                for(i=0; i<3;i++)
                        for(j=0;j<4;j++)
                                twoD[i][j] = i+j
            for(i=0; i<3;i++)
                {
                    for(j=0;j<4;j++)
                            System.out.print(twoD[i][j] + " ");
                     System.out.println();
                }
            }
    }
```

The output would be –

0 1 2 3

1 2 3 4

2 3 4 5

Program 1.13 Demonstration of irregular arrays

```
class UnevenArr
    {
        public static void main(String args[])
        {
                int twoD[][] = new int[3][];
                twoD[0] = new int[3];
                twoD[1] = new int[1];
                twoD[2] = new int[5];
                int i, j, k = 0;
                for(i=0; i<3;i++)
                    for(j=0;j< twoD[i].length;j++,k++)
                    twoD[i][j] = k;

                for(i=0; i<3;i++)
                {
                        for(j=0; j< twoD[i].length;j++)
                        System.out.print(twoD[i][j] + " ");
                System.out.println();
                }
            }
    }
```

The output would be –

```
0 1 2
3
```

4 5 6 7 8

Program2.1:Demonstration of left-shiftoperator

```
class ShiftDemo
{
        public static void main(String args[])
        {
                bytea=64,b; int i;
                i=a<<2;
                b = (byte)(a<< 2; )System.out.println("Originalvalueofa:"+a);
                System.out.println("i and b: " + i + "" + b);
        }
}
```

The result would be–

Original value of a: 64 i

and b: 256 0

Program 2.2

```
classShiftDemo1
{
        public static void main(String args[])
        {
                int i;
                int num=0xFFFFFFE;
                for(j=0;j<4;j++)
                {
                        num = num << 1;
```

```
                System.out.println(num);
            }
        }
}
```

The output would be–

536870908

1073741816 //twice the previous value

2147483632 //twice the previous value

-32 //crosses the range of int and hence negative

Program2.3Demonstration of Boolean Logical operators

```
Class BoolLogic
 {
        public static void main(String args[])
        {
                boolean a = true;
                boolean b=false;
                boolean c=a|b;
                boolean d=a&b;
```

```
                boolean e =a^b;
                boolean f=(!a&b)|(a&!b);
                boolean g = !a;
                System.out.println(" a = " + a);
                System.out.println(" b = " + b);
                System.out.println(" a|b = " + c);
                System.out.println(" a&b = " + d);
                System.out.println(" a^b = " + e);
                System.out.println("!a&b|a&!b="+f);
                System.out.println(" !a = " + g);
                boolean h=b&(a=!a);
                 System.out.println("b&(a=!a)="+h);
                System.out.println("New a is "+a);
            }
}
```

The output would be–

```
a=true
b = false
a|b = true
a&b=false
a^b=true
!a&b|a&!b=true
!a=false
b&(a=!a)=false
 New a is false
```

Program2.4Demonstration of for-each version of for loop

```
class ForEach
{
        public static void main(String args[])
        {
                int sum =0;
                int nums[][]=new int[2][3];

                 // give nums some values
                for(inti=0;i<2;i++)
                        for(intj=0;j<3;j++)
                        nums[i][j]=(i+1)*(j+1);

                for(intx[]:nums)  //numsisa2-darrayandxis1-darray
                {
                        for(inty:x)  //yreferselementsin1-darrayx
                        {
                         System.out.println("Valueis:"+y); sum += y;
                         }
                }
                         System.out.println("Summation:"+sum);
        }
}
```

The output would be–

```
Value is:1
Value is:2
Value is:3
Value is:2
```

Value is:4

Value is:6

Summation:18

Program2.5 Linear/SequentialSearch

```
class SeqSearch
{
        public static void main(String args[])
        {
                int nums[]={6,8,3,7,5,6,1,4};
                int val= 5;
                boolean found=false;
                for(int x:nums)
                 {
                        if(x==val)
                        {
                        found=true; break;
                        }
                }
                if(found)
                        System.out.println("Value found!");
              }
   }
```

The output would be–

Value found!

Program2.6 Illustration of break statement with labels

```
class Break
{
        public static void main(String args[])
        {
                boolean t=true;
                 first:
                second:
                {
                        third:
                        {
                                System.out.println("Before the break.");
                                if(t)
                                        break second;
                                System.out.println("This won't execute");
                        }
                                System.out.println("This won't execute");
            }
                System.out.println("This is after second block.");
        }
}
```

The output would be–

Before the break

This is after second block

Program that asks asks a user for their name and age, then prints message back to them.

```
import java.util.Scanner; // Step 1: Import the Scanner class
 public class Main {
    public static void main(String[] args) {
        // Step 2: Create a Scanner object
        Scanner input = new Scanner(System.in); // Prompt the user for text input
        System.out.print("Enter your name: ");
        String name = input.nextLine(); // Reads a line of text


            // Prompt the user for numeric input
        System.out.print("Enter your age: ");
        int age = input.nextInt(); // Reads an integer


        // Output the data provided by the user
        System.out.println("Hello, " + name + "! You are " + age + " years old.");


        // Step 4: Close the scanner
        input.close();
    }}
```